

MANCHESTER AP, United Kingdom

WMO: 033340

Lat: 53.354N

Lon: 2.275W

Elev: 78

StdP: 100.39

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.1	-6.7	2.2	-1.0	-5.2	2.5	0.0	13.7	7.7	12.3	7.9	1.5	90	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	25.8	18.1	23.8	17.2	21.9	16.4	18.8	23.5	17.9	22.3	17.1	20.8	3.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	20.6	16.2	11.6	19.5	15.4	11.1	18.7	53.5	23.7	50.8	22.4	48.2	20.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.6	8.5	DB	-6.8	28.6	2.3	2.0	-8.4	30.0	-9.8	31.2	-11.0	32.4	-12.7	33.8	
			WB	-7.2	20.8	2.3	1.3	-8.8	21.7	-10.1	22.4	-11.4	23.1	-13.0	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.6	4.9	6.5	8.8	11.8	14.5	16.5	16.2	13.8	10.8	7.2	4.9
	DBStd	5.18	3.06	3.06	2.85	2.90	2.95	2.57	2.57	2.41	2.34	2.82	3.17	3.63
	HDD10.0	776	168	143	114	57	14	1	0	0	1	24	92	161
	HDD18.3	3049	427	376	368	287	203	120	71	76	137	233	334	418
	CDD10.0	798	1	1	4	20	69	134	201	192	116	49	8	2
	CDD18.3	29	0	0	0	0	1	4	13	10	2	0	0	0
	CDH23.3	206	0	0	0	0	7	35	103	55	5	1	0	0
	CDH26.7	32	0	0	0	0	0	4	17	11	0	0	0	0
Wind	WSAvg	4.0	4.6	4.5	4.4	4.0	4.1	3.7	3.5	3.5	3.6	4.0	4.0	4.1
Precipitation	PrecAvg	832	72	56	57	55	60	66	65	83	76	86	85	83
	PrecMax	1073	137	161	139	108	144	182	152	185	180	194	181	200
	PrecMin	577	8	4	11	4	8	16	10	7	11	24	40	15
	PrecStd	114	31	35	28	25	29	34	33	38	37	40	33	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.9	16.1	21.0	24.1	27.1	28.8	28.3	23.9	19.2	15.2	13.8
		MCWB	10.9	10.5	11.0	13.8	16.3	18.5	18.9	18.6	17.4	15.7	13.3	11.9
	2%	DB	11.2	11.9	13.9	17.5	21.9	23.9	26.0	24.4	21.1	17.2	14.0	12.2
		MCWB	9.5	9.6	10.2	11.9	15.2	17.1	18.0	17.9	16.6	14.6	12.4	10.7
	5%	DB	10.1	10.6	12.2	15.3	19.3	21.7	24.0	22.1	19.2	16.1	12.9	11.1
		MCWB	8.6	8.7	9.3	10.8	13.8	15.9	17.5	17.0	15.2	13.9	11.2	9.7
	10%	DB	9.1	9.6	11.0	13.9	17.1	19.8	21.9	20.6	18.0	15.1	11.8	10.0
		MCWB	7.9	7.9	8.5	10.0	12.6	14.8	16.5	16.2	14.4	13.0	10.3	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.3	12.1	14.1	17.5	19.7	20.2	19.8	18.7	16.5	13.9	12.2
		MCDB	12.6	12.9	14.8	18.3	23.0	24.7	25.6	24.8	22.2	18.7	15.1	13.4
	2%	WB	10.0	10.0	10.8	12.7	15.6	17.7	18.8	18.6	17.3	15.3	12.7	11.2
		MCDB	11.0	11.2	13.1	16.7	20.4	22.3	24.1	22.9	20.2	17.0	13.9	12.4
	5%	WB	8.9	9.0	9.9	11.4	14.5	16.3	17.9	17.7	16.0	14.3	11.5	10.1
		MCDB	9.8	10.4	11.9	14.9	18.6	20.5	22.6	21.4	18.3	15.8	12.7	11.1
	10%	WB	7.9	8.0	8.9	10.3	13.2	15.3	17.0	16.9	15.2	13.2	10.5	8.8
		MCDB	9.0	9.3	10.9	13.3	16.4	18.8	21.2	19.7	17.4	14.7	11.6	9.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.6	6.8	7.8	8.1	8.0	8.1	7.6	7.2	6.2	5.3	5.0
		MCDBR	5.7	6.2	8.5	10.1	11.5	11.8	12.0	10.0	9.0	6.8	5.4	5.7
		MCWBR	5.3	4.8	5.7	5.9	6.3	6.2	5.5	4.9	5.3	4.8	4.7	5.2
	5% WB	MCDBR	5.5	5.9	7.6	9.4	10.3	10.2	10.1	8.6	7.8	6.3	5.3	5.5
		MCWBR	5.4	5.0	5.6	5.9	6.1	6.0	5.4	4.8	5.3	5.0	4.9	5.3
Clear-Sky Solar Irradiance	taub	0.325	0.339	0.369	0.395	0.398	0.397	0.397	0.382	0.367	0.349	0.324	0.313	
	taud	2.321	2.371	2.294	2.239	2.273	2.303	2.307	2.364	2.410	2.427	2.385	2.297	
	Ebn at Noon	637	759	807	833	851	854	846	838	805	738	638	571	
	Edh at Noon	60	78	103	123	125	122	120	107	91	73	56	51	
All-Sky Solar Radiation	RadAvg	0.65	1.33	2.46	3.81	4.83	5.05	4.80	3.95	2.85	1.59	0.79	0.50	
	RadStd	0.06	0.15	0.23	0.37	0.34	0.47	0.48	0.26	0.22	0.14	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page